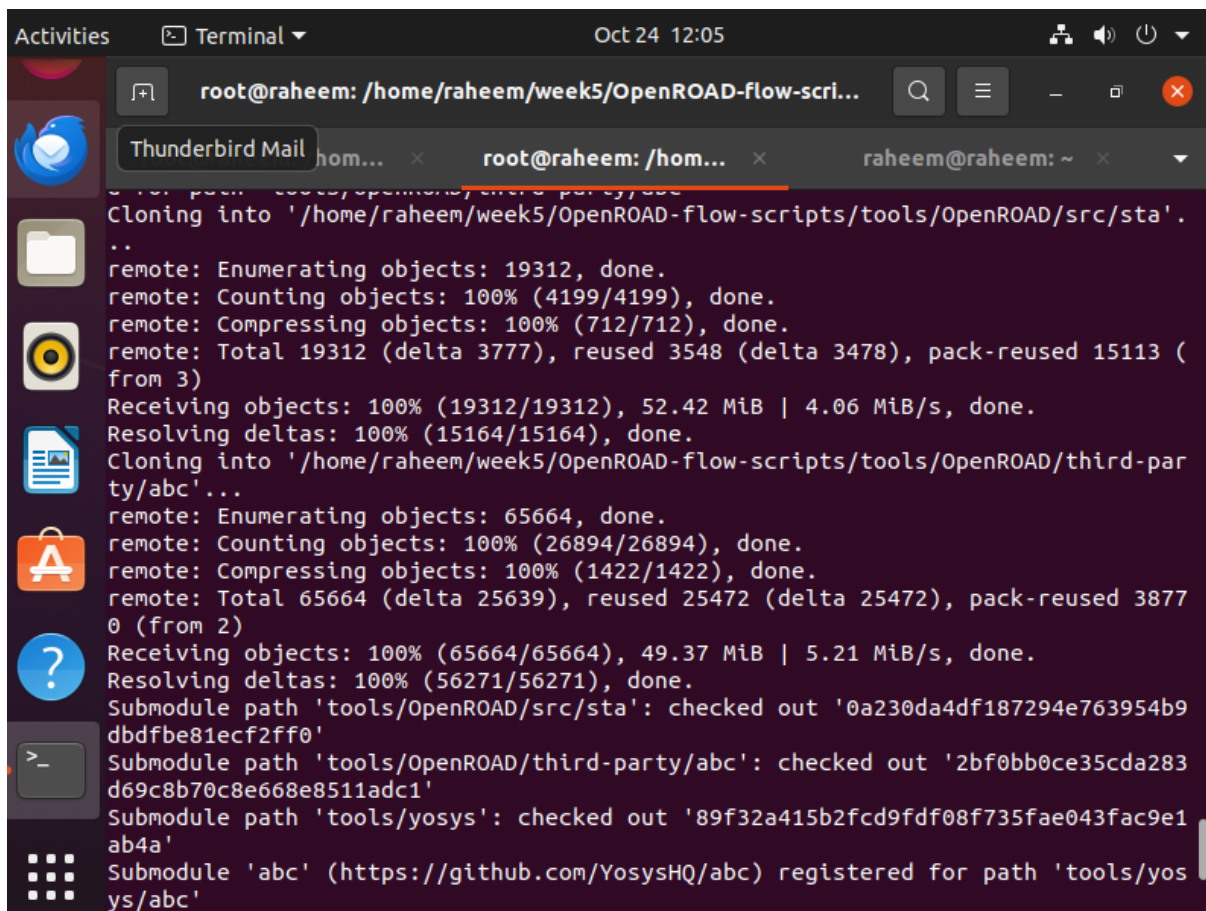


OpenROAD is an open-source, fully automated **RTL-to-GDSII** toolchain for **digital IC (Integrated Circuit) design**. It offers an end-to-end physical design flow — from register-transfer level (RTL) synthesis to final layout (GDSII) output — supporting synthesis, floorplanning, placement, clock tree synthesis, routing, and layout generation. Its modular, scriptable, and GUI-supported environment makes it suitable for both **academic research** and **industrial prototyping**



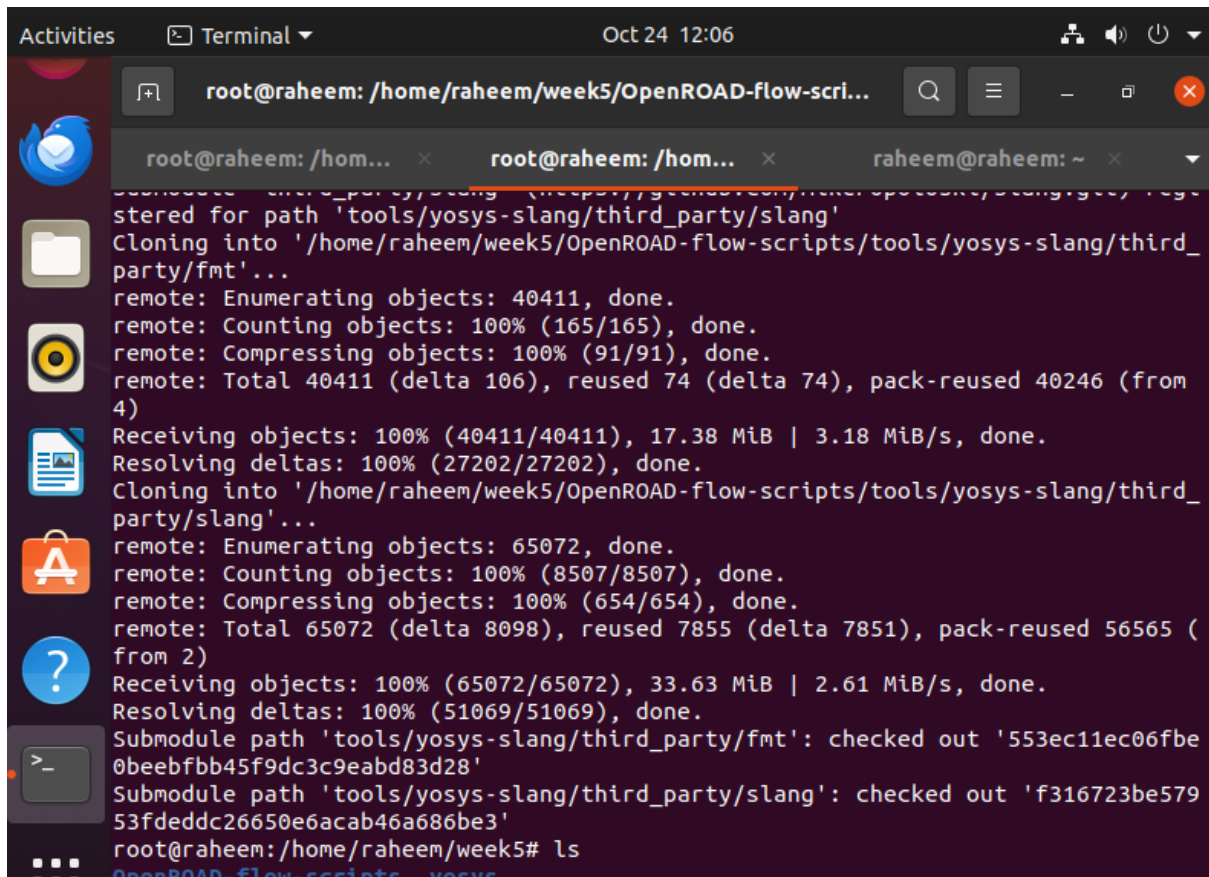
```
root@raheem: /home/raheem/week5/OpenROAD-flow-scri...
Cloning into '/home/raheem/week5/OpenROAD-flow-scripts/tools/OpenROAD/src/sta'.
..
remote: Enumerating objects: 19312, done.
remote: Counting objects: 100% (4199/4199), done.
remote: Compressing objects: 100% (712/712), done.
remote: Total 19312 (delta 3777), reused 3548 (delta 3478), pack-reused 15113 (
from 3)
Receiving objects: 100% (19312/19312), 52.42 MiB | 4.06 MiB/s, done.
Resolving deltas: 100% (15164/15164), done.
Cloning into '/home/raheem/week5/OpenROAD-flow-scripts/tools/OpenROAD/third-par
ty/abc'...
remote: Enumerating objects: 65664, done.
remote: Counting objects: 100% (26894/26894), done.
remote: Compressing objects: 100% (1422/1422), done.
remote: Total 65664 (delta 25639), reused 25472 (delta 25472), pack-reused 3877
0 (from 2)
Receiving objects: 100% (65664/65664), 49.37 MiB | 5.21 MiB/s, done.
Resolving deltas: 100% (56271/56271), done.
Submodule path 'tools/OpenROAD/src/sta': checked out '0a230da4df187294e763954b9
dbdfbe81ecf2ff0'
Submodule path 'tools/OpenROAD/third-party/abc': checked out '2bf0bb0ce35cda283
d69c8b70c8e668e8511adc1'
Submodule path 'tools/yosys': checked out '89f32a415b2fcd9fdf08f735fae043fac9e1
ab4a'
Submodule 'abc' (https://github.com/YosysHQ/abc) registered for path 'tools/yos
ys/abc'
```

Step 1:: Install Prerequisites

Update your system and install core build tools:

```
sudo apt update && sudo apt upgrade -y
```

```
sudo apt install -y build-essential cmake clang g++ gcc git python3 python3-dev \
libboost-all-dev libtcl tcl-dev tcllib libreadline-dev zlib1g-dev flex bison \
swig libpcre3-dev qtbase5-dev libliblemon-dev libspdlog-dev libeigen3-dev libffi-dev \
pkg-config libjson-c-dev libzstd-dev
```



The screenshot shows a terminal window with the following output:

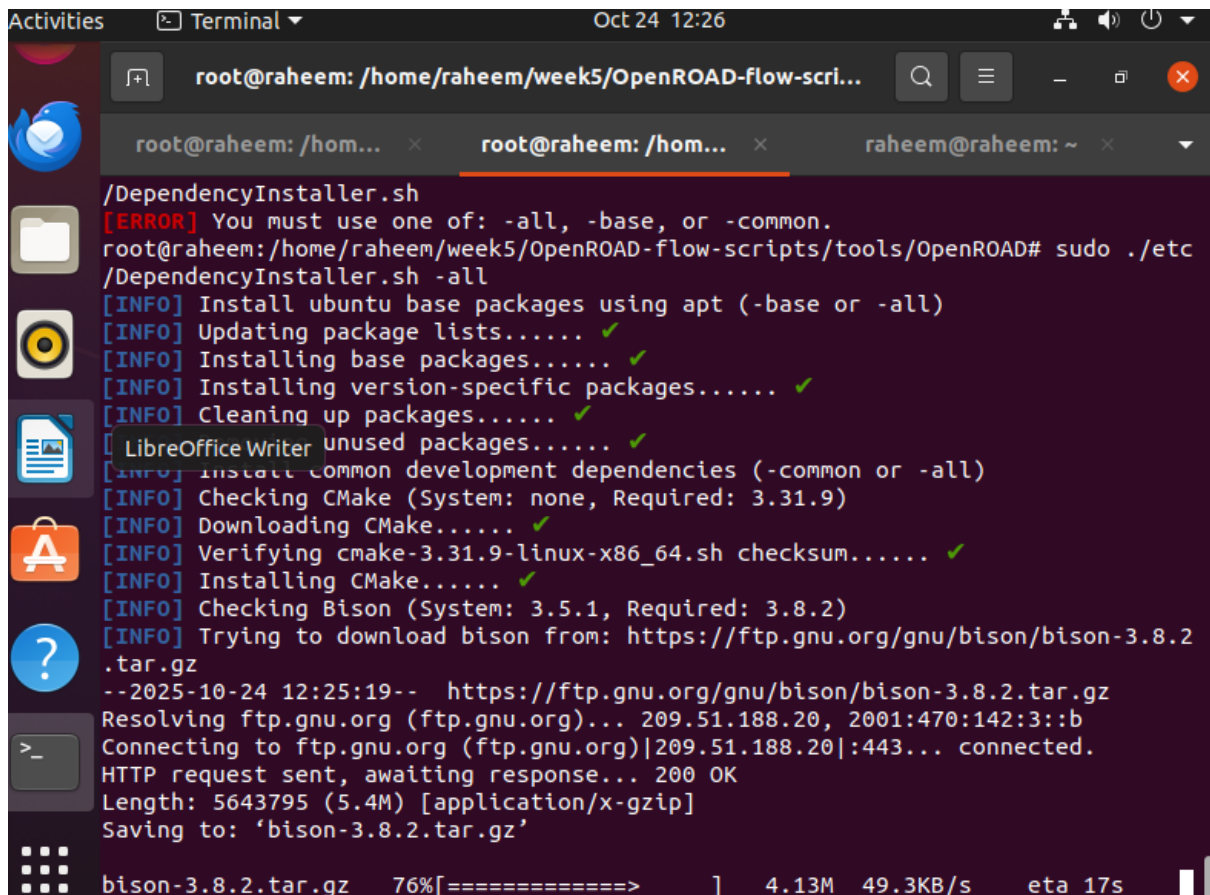
```
root@raheem: /home/raheem/week5/OpenROAD-flow-scri...
Cloning into '/home/raheem/week5/OpenROAD-flow-scripts/tools/yosys-slang/third_
party/fmt'...
remote: Enumerating objects: 40411, done.
remote: Counting objects: 100% (165/165), done.
remote: Compressing objects: 100% (91/91), done.
remote: Total 40411 (delta 106), reused 74 (delta 74), pack-reused 40246 (from
4)
Receiving objects: 100% (40411/40411), 17.38 MiB | 3.18 MiB/s, done.
Resolving deltas: 100% (27202/27202), done.
Cloning into '/home/raheem/week5/OpenROAD-flow-scripts/tools/yosys-slang/third_
party/slang'...
remote: Enumerating objects: 65072, done.
remote: Counting objects: 100% (8507/8507), done.
remote: Compressing objects: 100% (654/654), done.
remote: Total 65072 (delta 8098), reused 7855 (delta 7851), pack-reused 56565 (
from 2)
Receiving objects: 100% (65072/65072), 33.63 MiB | 2.61 MiB/s, done.
Resolving deltas: 100% (51069/51069), done.
Submodule path 'tools/yosys-slang/third_party/fmt': checked out '553ec11ec06fbe
0beebfbb45f9dc3c9eabd83d28'
Submodule path 'tools/yosys-slang/third_party/slang': checked out 'f316723be579
53fdeddc26650e6acab46a686be3'
root@raheem: /home/raheem/week5# ls
OpenROAD-flow-scripts yosys
```

Step 2: Clone the Repositories

Install OpenROAD-flow scripts (wrapper for Yosys, OpenROAD, etc.)

```
git clone --recursive https://github.com/The-OpenROAD-Project/OpenROAD-flow-scripts.git
```

```
cd OpenROAD-flow-scripts
```

A terminal window titled 'Terminal' with a timestamp of 'Oct 24 12:26'. The window shows a root user at a machine named 'raheem' in the directory '/home/raheem/week5/OpenROAD-flow-scri...'. The user has executed the command 'sudo ./etc/DependencyInstaller.sh -all'. The script output shows various installation steps: updating package lists, installing base and version-specific packages, cleaning up, installing common development dependencies, checking and installing CMake, and downloading bison. The bison download progress is shown at the bottom as 76% complete.

```
root@raheem: /home/raheem/week5/OpenROAD-flow-scri...
/DependencyInstaller.sh
[ERROR] You must use one of: -all, -base, or -common.
root@raheem: /home/raheem/week5/OpenROAD-flow-scripts/tools/OpenROAD# sudo ./etc
/DependencyInstaller.sh -all
[INFO] Install ubuntu base packages using apt (-base or -all)
[INFO] Updating package lists..... ✓
[INFO] Installing base packages..... ✓
[INFO] Installing version-specific packages..... ✓
[INFO] Cleaning up packages..... ✓
[INFO] LibreOffice Writer unused packages..... ✓
[INFO] Install common development dependencies (-common or -all)
[INFO] Checking CMake (System: none, Required: 3.31.9)
[INFO] Downloading CMake..... ✓
[INFO] Verifying cmake-3.31.9-linux-x86_64.sh checksum..... ✓
[INFO] Installing CMake..... ✓
[INFO] Checking Bison (System: 3.5.1, Required: 3.8.2)
[INFO] Trying to download bison from: https://ftp.gnu.org/gnu/bison/bison-3.8.2
.tar.gz
--2025-10-24 12:25:19-- https://ftp.gnu.org/gnu/bison/bison-3.8.2.tar.gz
Resolving ftp.gnu.org (ftp.gnu.org)... 209.51.188.20, 2001:470:142:3::b
Connecting to ftp.gnu.org (ftp.gnu.org)|209.51.188.20|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 5643795 (5.4M) [application/x-gzip]
Saving to: 'bison-3.8.2.tar.gz'
bison-3.8.2.tar.gz 76%[=====> ] 4.13M 49.3KB/s eta 17s
```

Step 3: Install and Build Dependencies

`sudo ./etc/DependencyInstaller.sh`

```
Activities Terminal Oct 24 12:27
root@raheem: /home/raheem/week5/OpenROAD-flow-scri...

root@raheem: /hom... x root@raheem: /hom... x raheem@raheem: ~ x

Saving to: 'bison-3.8.2.tar.gz'
bison-3.8.2.tar.gz 100%[=====] 5.38M 101KB/s in 79s
2025-10-24 12:26:39 (69.9 KB/s) - 'bison-3.8.2.tar.gz' saved [5643795/5643795]

[INFO] Verifying bison-3.8.2.tar.gz checksum..... ✓
[INFO] Extracting Bison..... ✓
[INFO] Configuring Bison..... ✓
[INFO] Building and installing Bison..... ✓
[INFO] Checking Flex (System: 2.6.4, Required: 2.6.4)
[INFO] Checking PCRE (System: none, Required: 10.42)
[INFO] Downloading PCRE..... ✓
[INFO] Verifying pcre2-10.42.tar.gz checksum..... ✓
[INFO] Extracting PCRE..... ✓
[INFO] Configuring PCRE..... ✓
[INFO] Building PCRE..... ✓
[INFO] Installing PCRE..... ✓
[INFO] Checking SWIG (System: none, Required: 4.3.0)
[INFO] Downloading SWIG..... ✓
[INFO] Verifying v4.3.0.tar.gz checksum..... ✓
[INFO] Extracting SWIG..... ✓
[INFO] Checking PCRE (System: 10.42, Required: 10.42)
[INFO] Generating SWIG configure script..... ✓
[INFO] Configuring SWIG..... ✓
```

```
Activities Terminal Oct 24 12:32
root@raheem: /home/raheem/week5/OpenROAD-flow-scri...

root@raheem: /hom... x root@raheem: /hom... x raheem@raheem: ~ x

[INFO] Checking or-tools (System: none, Required: 9.11.4210)
[INFO] Downloading or-tools..... ✓
[INFO] Extracting or-tools..... ✓

[INFO] Installation Summary
=====
====
Package | System Version | Required Version | Status
-----|-----|-----|-----
CMake | none | 3.31.9 | installed
Bison | 3.5.1 | 3.8.2 | installed
Flex | 2.6.4 | 2.6.4 | skipped
PCRE | none | 10.42 | installed
S Ubuntu Software | none | 4.3.0 | installed
Boost | none | 1.86 | installed
Eigen | none | 3.4 | installed
CUDD | none | 3.0.0 | installed
CUSP | none | any | installed
Lemon | none | 1.3.1 | installed
spdlog | none | 1.15.0 | installed
gtest | none | 1.17.0 | installed
Abseil | none | 20240722 | installed
or-tools | none | 9.11.4210 | installed
=====
root@raheem: /home/raheem/week5/OpenROAD-flow-scripts/tools/OpenROAD#
```

```
Activities Terminal Oct 24 14:08
root@raheem: /home/raheem/week5/OpenROAD-flow-scri...
root@raheem: /hom... root@raheem: /hom... raheem@raheem: ~
(OSS CAD Suite) root@raheem: /home/raheem/week5/OpenROAD-flow-scripts# ./etc/Dep
endencyInstaller.sh -all
[INFO] Install ubuntu base packages using apt (-base or -all)
[INFO] Updating package lists..... ✓
[INFO] Installing base packages..... ✓
[INFO] Installing version-specific packages..... ✓
[INFO] Cleaning up packages..... ✓
[INFO] Removing unused packages..... ✓
[INFO] Install common development dependencies (-common or -all)
[INFO] Checking CMake (System: 3.31.9, Required: 3.31.9)
[INFO] Checking Bison (System: 3.8.2, Required: 3.8.2)
[INFO] Checking Flex (System: 2.6.4, Required: 2.6.4)
[INFO] Checking PCRE (System: 10.42, Required: 10.42)
[INFO] Checking SWIG (System: 4.3.0, Required: 4.3.0)
[INFO] Checking Boost (System: 1.86, Required: 1.86)
[INFO] Checking Eigen (System: 3.4, Required: 3.4)
[INFO] Checking CUDD (Required: 3.0.0)
[INFO] Checking CUSP
[INFO] Checking Lemon (System: 1.3.1, Required: 1.3.1)
[INFO] Checking spdlog (System: 1.15.0, Required: 1.15.0)
[INFO] Checking gtest (Required: 1.17.0)
[INFO] Checking Abseil (System: 20240722, Required: 20240722)
[INFO] Install equivalence dependencies (-eqy)...
[INFO] Checking Yosys (System: 0.58+84, Required: 0.58)
[INFO] Cloning Yosys v0.58..... ✓
[INFO] Building Yosys.....
```

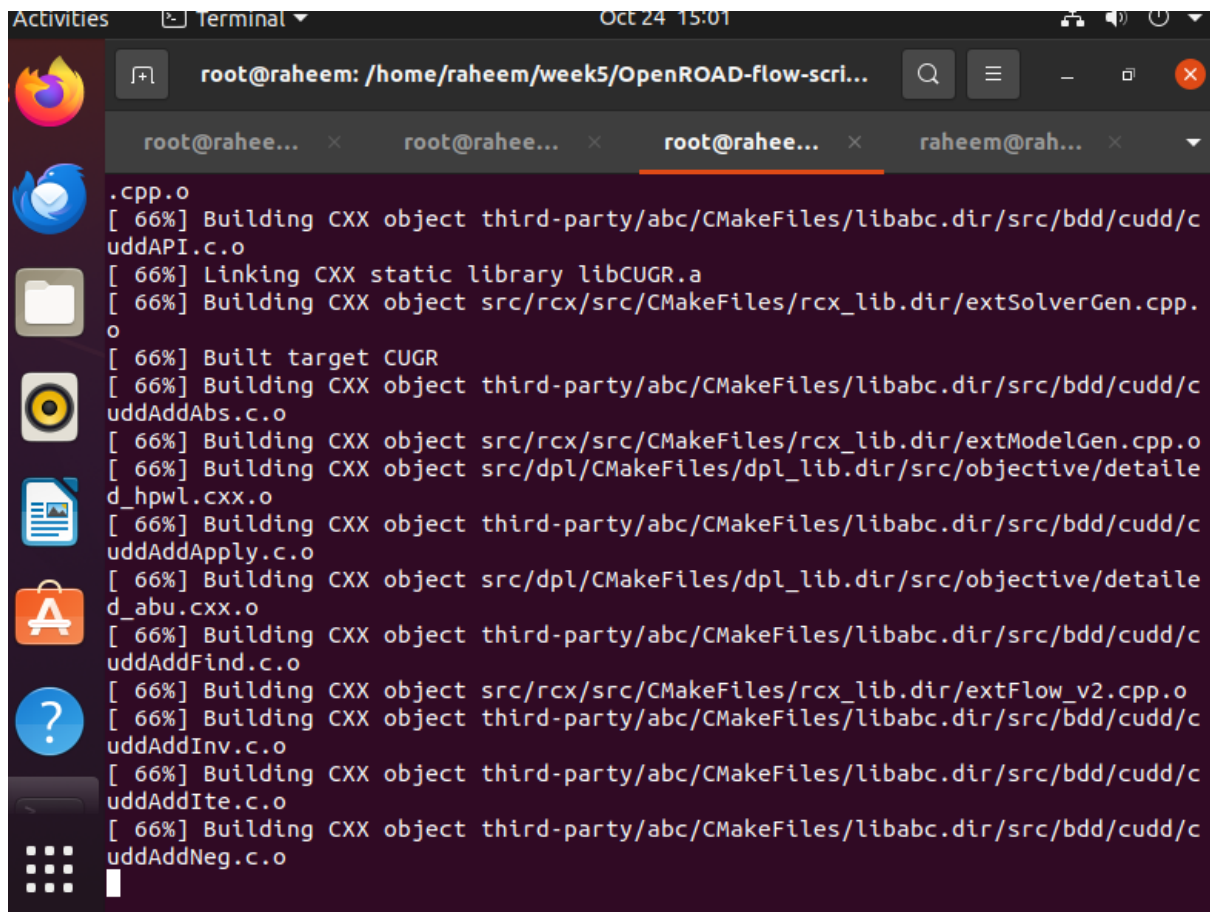
```
Activities Terminal Oct 24 14:20
root@raheem: /home/raheem/week5/OpenROAD-flow-scri...
root@raheem: /hom... root@raheem: /hom... root@raheem: /hom...
```

Package	System Version	Required Version	Status
CMake	3.31.9	3.31.9	skipped
Bison	3.8.2	3.8.2	skipped
Flex	2.6.4	2.6.4	skipped
PCRE	10.42	10.42	skipped
SWIG	4.3.0	4.3.0	skipped
Boost	1.86	1.86	skipped
Eigen	3.4	3.4	skipped
CUDD	found	3.0.0	skipped
CUSP	found	any	skipped
Lemon	1.3.1	1.3.1	skipped
spdlog	1.15.0	1.15.0	skipped
gtest	found	1.17.0	skipped
Abseil	20240722	20240722	skipped
Yosys	0.58+84	0.58	installed
Eqy	v0.58	v0.58	skipped
Sby	v0.58-2-g11fd202	v0.58	installed
or-tools	9.11.4210	9.11.4210	skipped

Step 4: Build OpenROAD Locally

Now build OpenROAD itself using the automated build script:

```
./build_openroad.sh --local
```



The screenshot shows a terminal window titled "Terminal" with the date and time "Oct 24 15:01". The terminal is running the command `./build_openroad.sh --local`. The output shows the progress of building various components, with each line starting with "[66%]". The components being built include:

- `third-party/abc/CMakeFiles/libabc.dir/src/bdd/cudd/cuddAPI.c.o`
- `src/rcx/src/CMakeFiles/rcx_lib.dir/extSolverGen.cpp.o`
- `src/rcx/src/CMakeFiles/rcx_lib.dir/extModelGen.cpp.o`
- `src/dpl/CMakeFiles/dpl_lib.dir/src/objective/detail_d_hpw.cxx.o`
- `third-party/abc/CMakeFiles/libabc.dir/src/bdd/cudd/cuddAddApply.c.o`
- `src/dpl/CMakeFiles/dpl_lib.dir/src/objective/detail_d_abu.cxx.o`
- `third-party/abc/CMakeFiles/libabc.dir/src/bdd/cudd/cuddAddFind.c.o`
- `src/rcx/src/CMakeFiles/rcx_lib.dir/extFlow_v2.cpp.o`
- `third-party/abc/CMakeFiles/libabc.dir/src/bdd/cudd/cuddAddInv.c.o`
- `third-party/abc/CMakeFiles/libabc.dir/src/bdd/cudd/cuddAddIte.c.o`
- `third-party/abc/CMakeFiles/libabc.dir/src/bdd/cudd/cuddAddNeg.c.o`

The terminal window also shows the system's activity bar on the left with icons for Firefox, Thunderbird, Files, Music, Documents, and Applications.

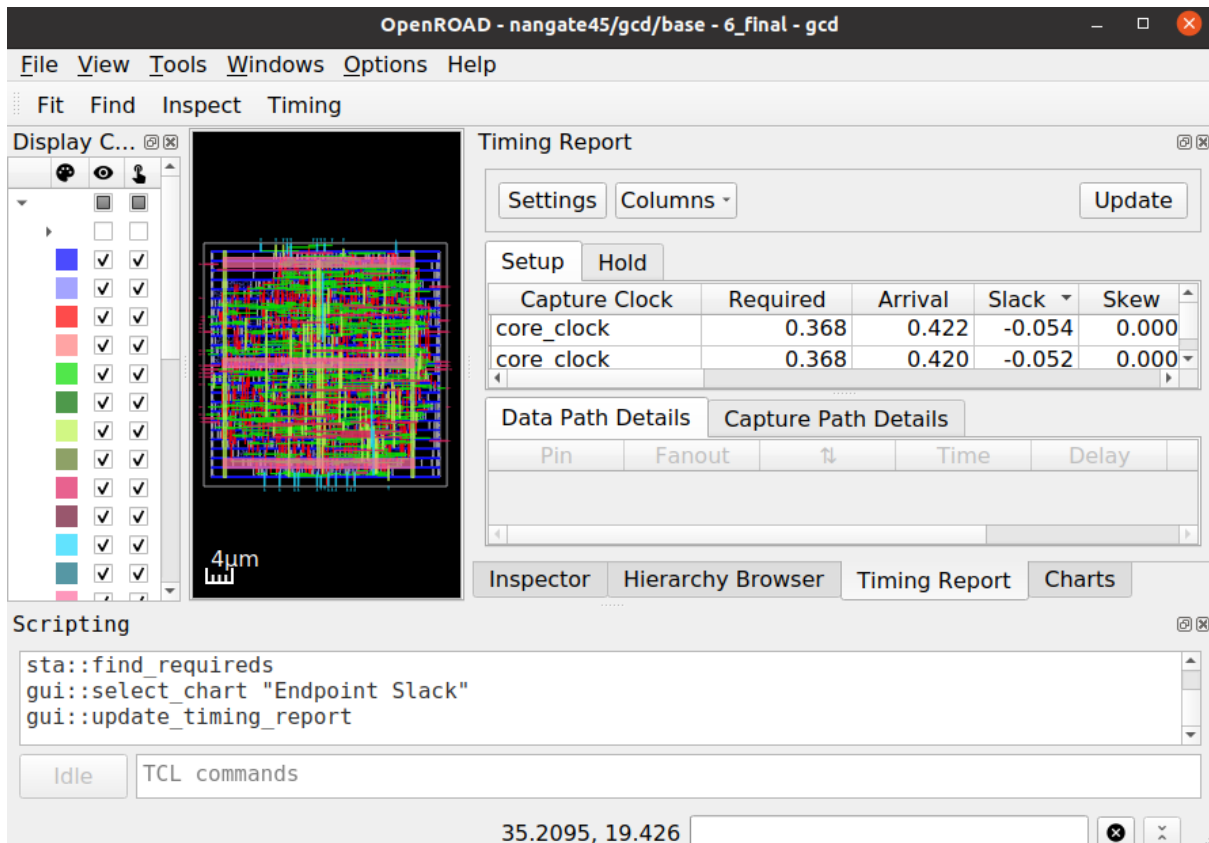
```
Oct 25 17:04
root@raheem: /home/raheem/week5/OpenROAD-flow-scri...
openroad: command not found
root@raheem:/home/raheem/week5/OpenROAD-flow-scripts# source ./env.sh
OPENROAD: /home/raheem/week5/OpenROAD-flow-scripts/tools/OpenROAD
root@raheem:/home/raheem/week5/OpenROAD-flow-scripts# openroad -V
OpenROAD v2.0-25608-gc8a6603b7f
Features included (+) or not (-): +GPU +GUI +Python
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
Usage: openroad [-help] [-version] [-no_init] [-no_splash] [-exit] [-gui] [-threads count|max] [-log file_name] [-metrics file_name] [-db file_name] [-no_settings] [-minimize] cmd_file
    -help                show help and exit
    -version              show version and exit
    -no_init              do not read .openroad init file
    -threads count|max   use count threads
    -no_splash            do not show the license splash at startup
    -exit                 exit after reading cmd_file
    -gui                  start in gui mode
    -minimize             start the gui minimized
    -no_settings          do not load the previous gui settings
    -python               start with python interpreter [limited to db operations]
    -log <file_name>     write a log in <file_name>
    -metrics <file_name> write metrics in <file_name> in JSON format
    -db <file_name>      open a .odb database at startup
    -no_settings          source cmd_file
root@raheem:/home/raheem/week5/OpenROAD-flow-scripts#
```

```
Activities Terminal Oct 25 17:06
root@raheem: /home/raheem/week5/OpenROAD-flow-scri...

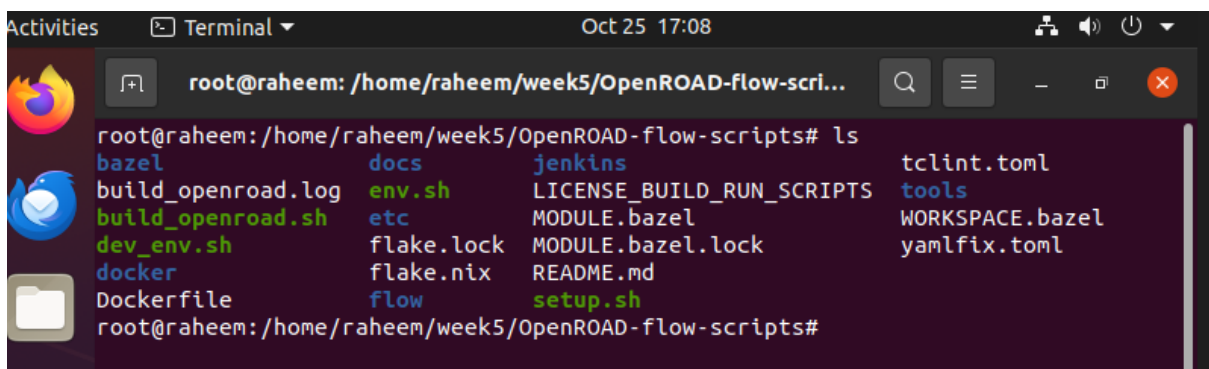
[INFO] GDS_ALLOW_EMPTY=fakeram.*
[INFO] All LEF cells have matching GDS/OAS cells
[INFO] No orphan cells in the final layout
Elapsed time: 0:01.43[h:]min:sec. CPU time: user 1.08 sys 0.16 (87%). Peak memo
ry: 434340KB.
cp results/nangate45/gcd/base/6_1_merged.gds results/nangate45/gcd/base/6_final
.gds
./logs/nangate45/gcd/base
Log
Elapsed/s Peak Memory/MB sha1sum .odb [0:20)
1_1_yosys_canonicalize 0 41 43d6c4de5f529681224d
1_2_yosys 0 44 d92ebd3244d38b043c3b
2_1_floorplan 0 120 cfb52234f67aefff5f2f
2_2_floorplan_macro 0 96 cfb52234f67aefff5f2f
2_3_floorplan_tapcell 0 96 e4dfd62a473fce5a7980
2_4_floorplan_pdn 0 98 89aa1f24b36071580d5e
3_1_place_gp_skip_io 0 99 0a7c4be117d31668c30b
3_2_place_iop 0 97 b39c8a1d1fcc15d1a952
3_3_place_gp 1 212 6178c75e809ff4504cd2
3_4_place_resized 0 118 6178c75e809ff4504cd2
3_5_place_dp 0 103 4e7dd0d4dfdd55393412
4_1_cts 3 126 cad6b04f2e308b46041c
5_1_grt 6 215 f313f5b83a406f22400a
5_2_route 11 1323 34de7c4d00358f64568a
5_3_fillcell 0 98 ffb82d8437e8c1a9ae45
6_1_fill 0 97 ffb82d8437e8c1a9ae45
6_1_merge 1 424 N/A
6_report 2 207 N/A
Total 24 1323
root@raheem: /home/raheem/week5/OpenROAD-flow-scripts/flow#
```

```
Activities Terminal Oct 25 17:07
root@raheem: /home/raheem/week5/OpenROAD-flow-scri...

[INFO] GDS_ALLOW_EMPTY=fakeram.*
[INFO] All LEF cells have matching GDS/OAS cells
[INFO] No orphan cells in the final layout
Elapsed time: 0:01.43[h:]min:sec. CPU time: user 1.08 sys 0.16 (87%). Peak memo
ry: 434340KB.
cp results/nangate45/gcd/base/6_1_merged.gds results/nangate45/gcd/base/6_final
.gds
./logs/nangate45/gcd/base
Log
Elapsed/s Peak Memory/MB sha1sum .odb [0:20)
1_1_yosys_canonicalize 0 41 43d6c4de5f529681224d
1_2_yosys 0 44 d92ebd3244d38b043c3b
2_1_floorplan 0 120 cfb52234f67aefff5f2f
2_2_floorplan_macro 0 96 cfb52234f67aefff5f2f
2_3_floorplan_tapcell 0 96 e4dfd62a473fce5a7980
2_4_floorplan_pdn 0 98 89aa1f24b36071580d5e
3_1_place_gp_skip_io 0 99 0a7c4be117d31668c30b
3_2_place_iop 0 97 b39c8a1d1fcc15d1a952
3_3_place_gp 1 212 6178c75e809ff4504cd2
3_4_place_resized 0 118 6178c75e809ff4504cd2
3_5_place_dp 0 103 4e7dd0d4dfdd55393412
4_1_cts 3 126 cad6b04f2e308b46041c
5_1_grt 6 215 f313f5b83a406f22400a
5_2_route 11 1323 34de7c4d00358f64568a
5_3_fillcell 0 98 ffb82d8437e8c1a9ae45
6_1_fill 0 97 ffb82d8437e8c1a9ae45
6_1_merge 1 424 N/A
6_report 2 207 N/A
Total 24 1323
root@raheem: /home/raheem/week5/OpenROAD-flow-scripts/flow# make gui flow
```

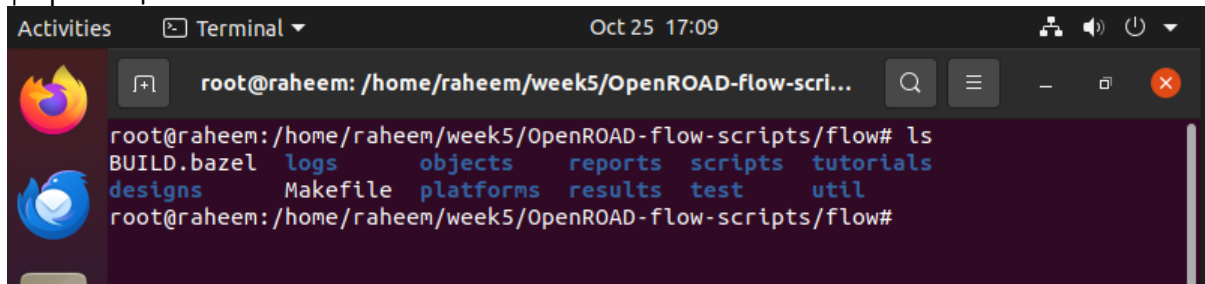



- └─ OpenROAD-flow-scripts
 - | └─ docker -> It has Docker based installation, run scripts and all saved here
 - | └─ docs -> Documentation for OpenROAD or its flow scripts.
 - | └─ flow -> Files related to run RTL to GDS flow
 - | └─ jenkins -> It contains the regression test designed for each build update
 - | └─ tools -> It contains all the required tools to run RTL to GDS flow
 - | └─ etc -> Has the dependency installer script and other things
 - | └─ setup_env.sh -> Its the source file to source all our OpenROAD rules to run the RTL to GDS flow



- └─ flow
 - | └─ design -> It has built-in examples from RTL to GDS flow across different technology nodes
 - | └─ makefile -> The automated flow runs through makefile setup
 - | └─ platform -> It has different technology note libraries, lef files, GDS etc

```
| | tutorials
| | util
| | scripts
```



The screenshot shows a terminal window titled "Terminal" with the date and time "Oct 25 17:09". The prompt is "root@raheem: /home/raheem/week5/OpenROAD-flow-scri...". The user has entered the command "ls" in the directory "/home/raheem/week5/OpenROAD-flow-scripts/flow#". The output lists the following files and directories: BUILD.bazel, logs, objects, reports, scripts, tutorials, designs, Makefile, platforms, results, test, and util.

```
root@raheem: /home/raheem/week5/OpenROAD-flow-scri...
root@raheem: /home/raheem/week5/OpenROAD-flow-scripts/flow# ls
BUILD.bazel  logs        objects     reports     scripts     tutorials
designs       Makefile    platforms   results     test        util
root@raheem: /home/raheem/week5/OpenROAD-flow-scripts/flow#
```

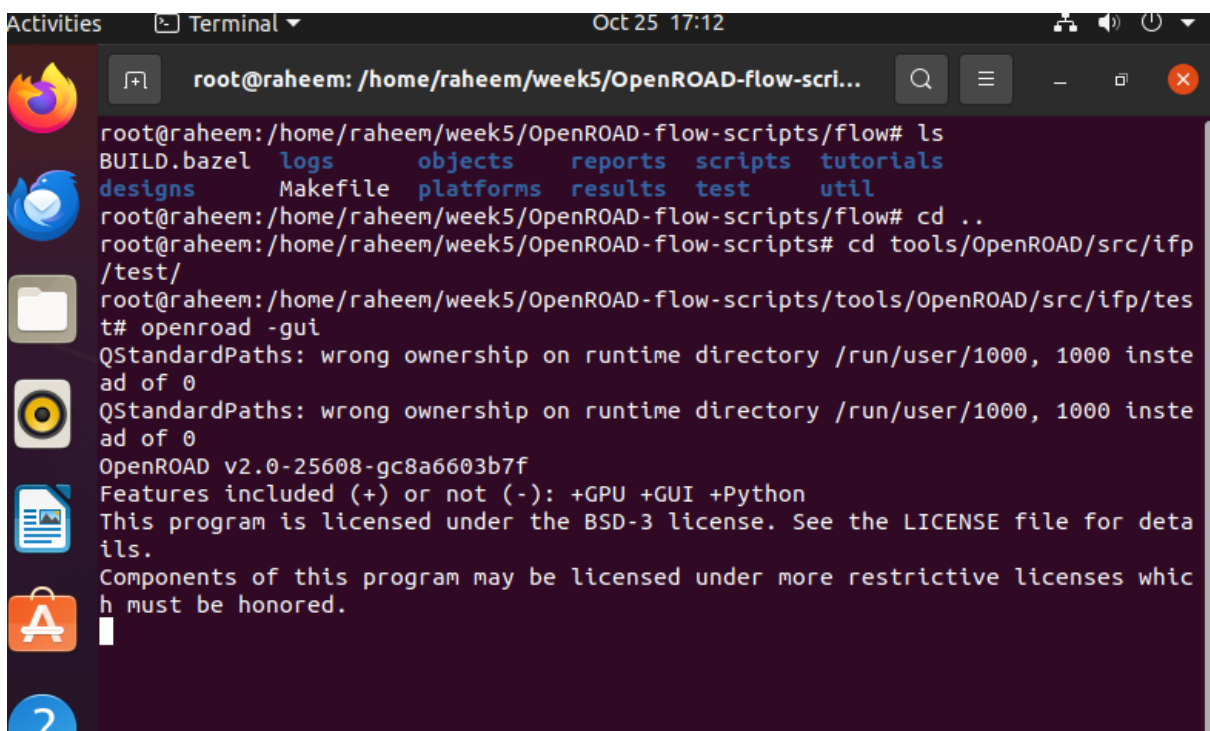
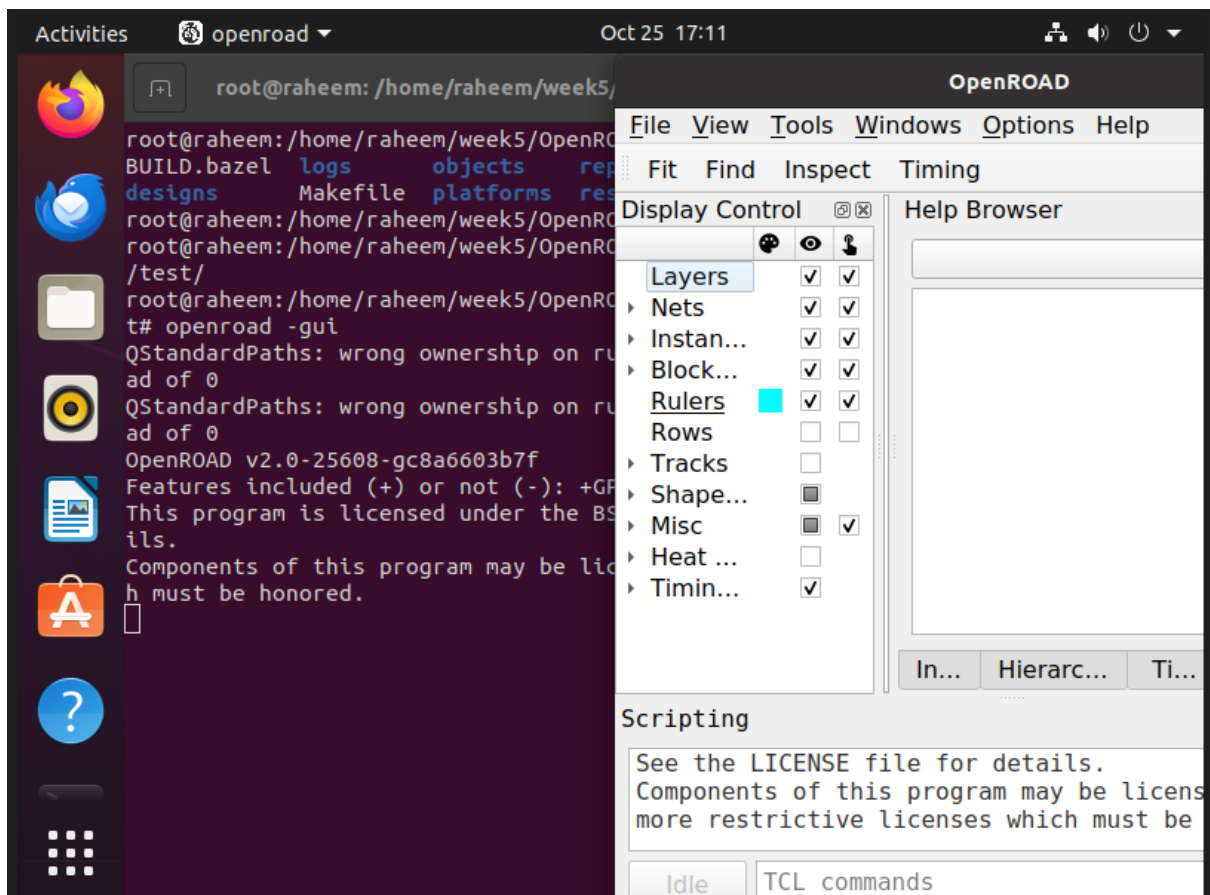
Floorplan Initialization Based On Core And Die Area

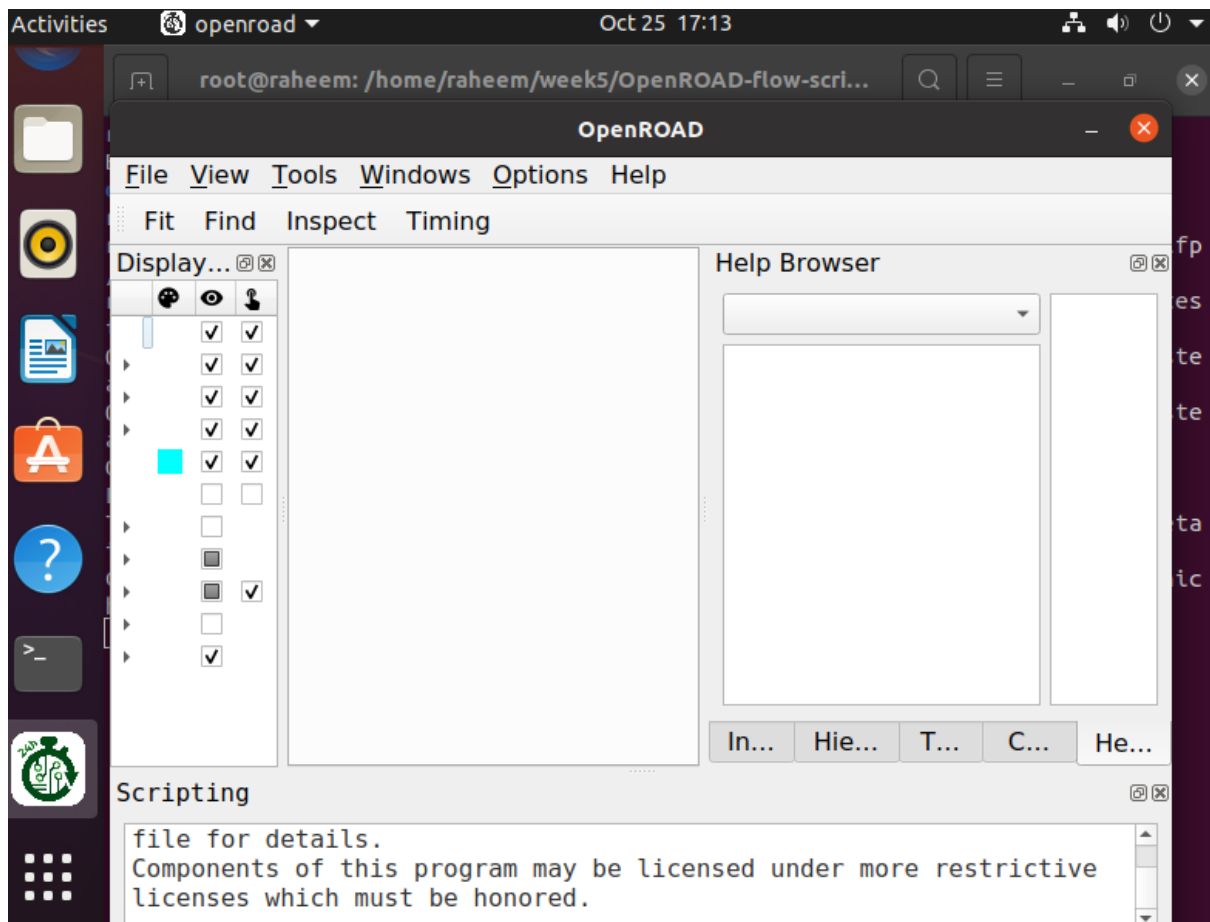
```
# init_floorplan
source "helpers.tcl"
read_lef Nangate45/Nangate45.lef
read_liberty Nangate45/Nangate45_typ.lib
read_verilog reg1.v
link_design top
initialize_floorplan -die_area "0 0 1000 1000" \
  -core_area "100 100 900 900" \
  -site FreePDK45_38x28_10R_NP_162NW_34O

set def_file [make_result_file init_floorplan1.def]
write_def $def_file
diff_files init_floorplan1.defok $def_file

cd ../tools/OpenROAD/src/ifp/test/

openroad -gui
```

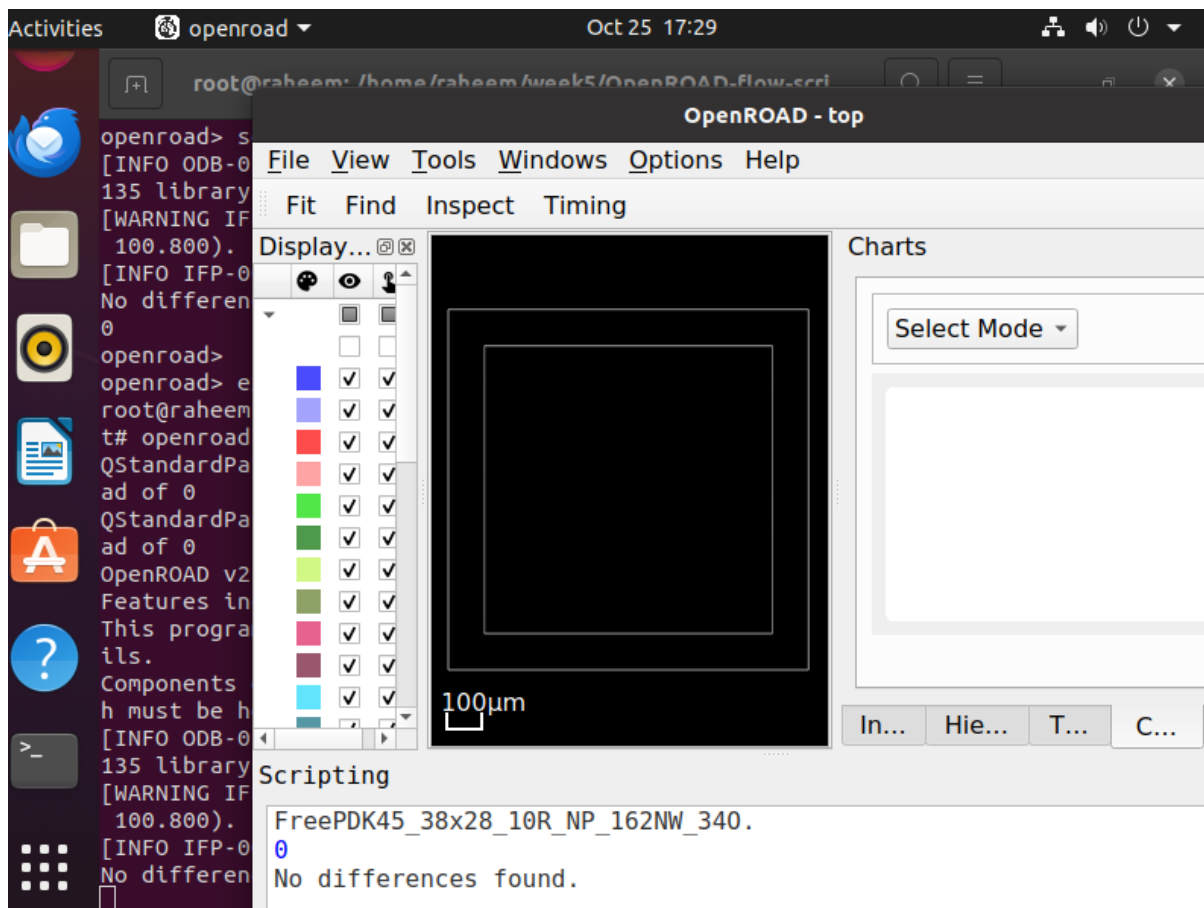




In Tcl Commands section GUI:

source init_floorplan1.tcl

```
openroad> source init_floorplan1.tcl
[INFO ODB-0227] LEF file: Nangate45/Nangate45.lef, created 22 layers, 27 vias,
135 library cells
[WARNING IFP-0028] Core area lower left (100.000, 100.000) snapped to (100.130,
100.800).
[INFO IFP-0001] Added 570 rows of 4209 site FreePDK45_38x28_10R_NP_162NW_340.
No differences found.
0
```



```

# init_floorplan -core_space < site height
source "helpers.tcl"
read_lef Nangate45/Nangate45.lef
read_liberty Nangate45/Nangate45_typ.lib
read_verilog reg1.v
link_design top
initialize_floorplan -utilization 30 \
  -aspect_ratio 0.5 \
  -core_space 1 \
  -site FreePDK45_38x28_10R_NP_162NW_340

```

```

set def_file [make_result_file init_floorplan2.def]
write_def $def_file
diff_files init_floorplan2.defok $def_file

```

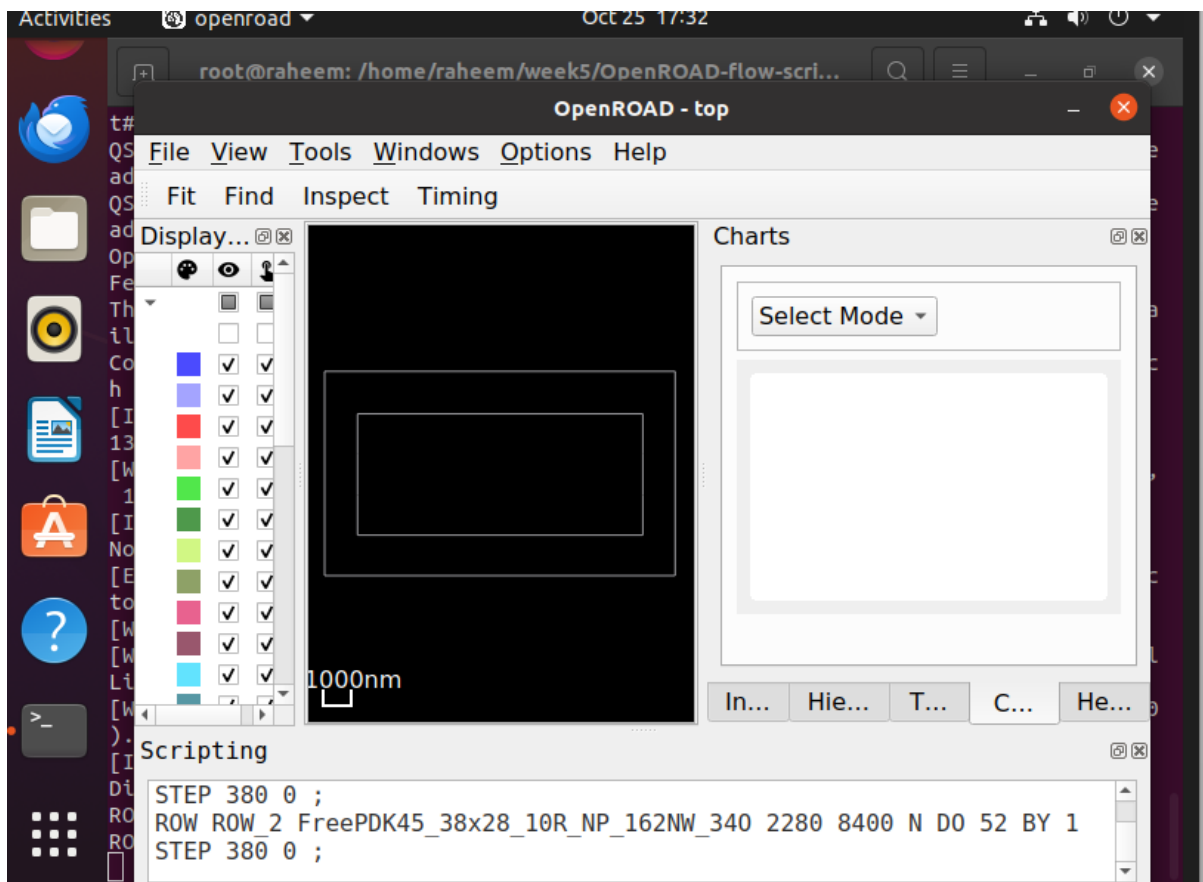
Run the following commands in the terminal in OpenROAD tool root directory to view how the floorplan initialized:

```
cd ../tools/OpenROAD/src/ifp/test/
```

```
openroad -gui
```

In the Tcl Commands section of the GUI:

```
source init_floorplan2.tcl
```

```
[WARNING STA-1140] Nangate45/Nangate45_typ.lib line 37, library NangateOpenCell
Library already exists.
[WARNING IFP-0028] Core area lower left (1.000, 1.000) snapped to (1.140, 1.400
).
[INFO IFP-0001] Added 3 rows of 52 site FreePDK45_38x28_10R_NP_162NW_340.
Differences found at line 7.
ROW ROW_0 FreePDK45_38x28_10R_NP_162NW_340 2280 2800 N D0 52 BY 1 STEP 380 0 ;
ROW ROW_2 FreePDK45_38x28_10R_NP_162NW_340 2280 8400 N D0 52 BY 1 STEP 380 0 ;
```

IO Pin Placement

Place pins on the boundary of the die on the track grid to minimize net wirelengths. Pin placement also creates a metal shape for each pin using min-area rules.

For designs with unplaced cells, the net wirelength is computed considering the center of the die area as the unplaced cells position.

Launch openroad GUI by running the following command(s) in the terminal in OpenROAD tool root directory:

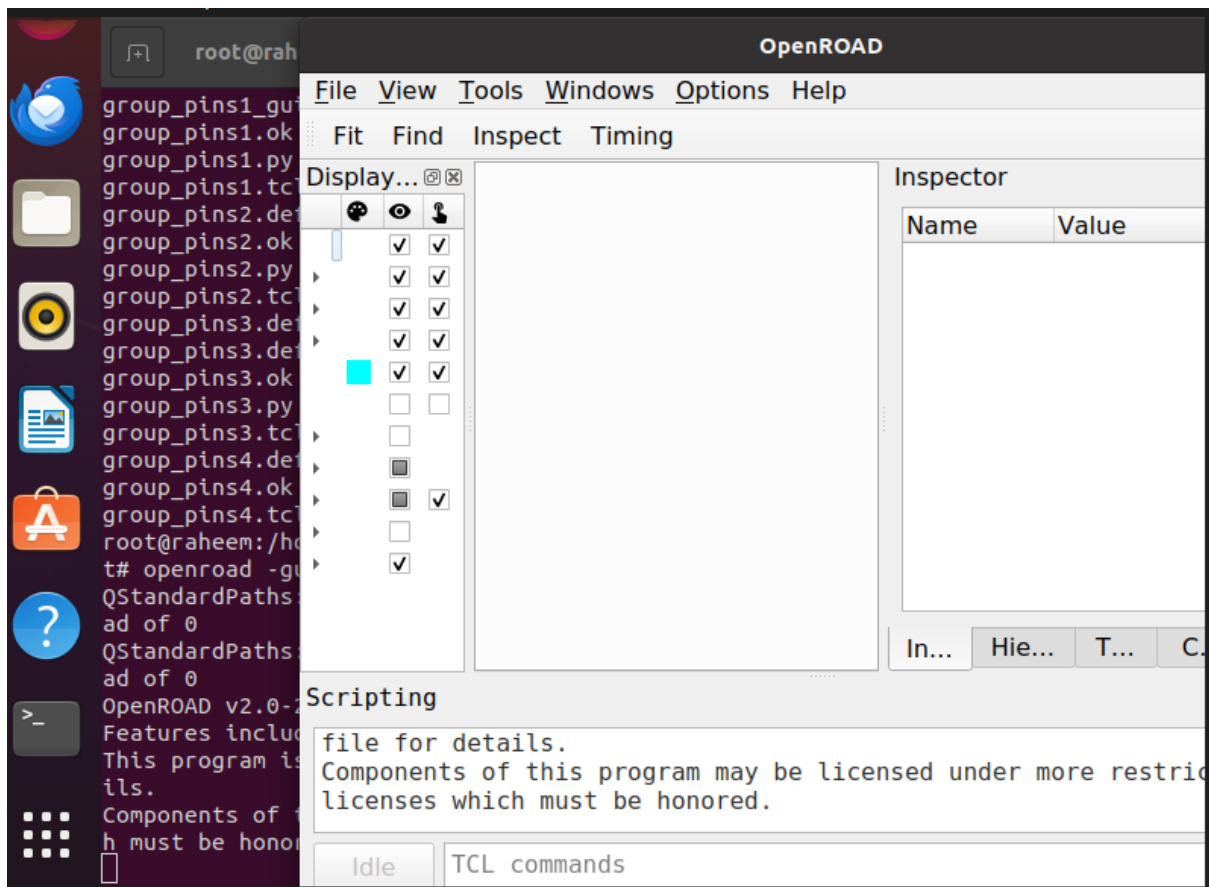
```
cd ../tools/OpenROAD/src/ppl/test/
```

```

root@raheem:/home/raheem/week5/OpenROAD-flow-scripts/tools/OpenROAD/src/tp# cd
..
root@raheem:/home/raheem/week5/OpenROAD-flow-scripts/tools/OpenROAD/src# ls
ant          dpl          gui          pad          sta
cgt          drt          ifp          par          stt
cmake        dst          Main.cc      pdn          tap
CMakeLists.txt est          Metrics.tcl  ppl          Tech.cc
cts          exa          mpl          psm          Timing.cc
cut          Exception.i  odb          ram          tst
dbSta        Exception-py.i OpenRoad.cc  rcx          upf
Design.cc    fin          OpenRoad.i  README.md    utl
Design.i     gpl          OpenRoad-py.i rmp
dft          grt          OpenRoad.tcl rsz
root@raheem:/home/raheem/week5/OpenROAD-flow-scripts/tools/OpenROAD/src#

```

openroad -gui



```

# gcd_nangate45 IO placement
source "helpers.tcl"
read_lef Nangate45/Nangate45.lef
read_def gcd.def

```

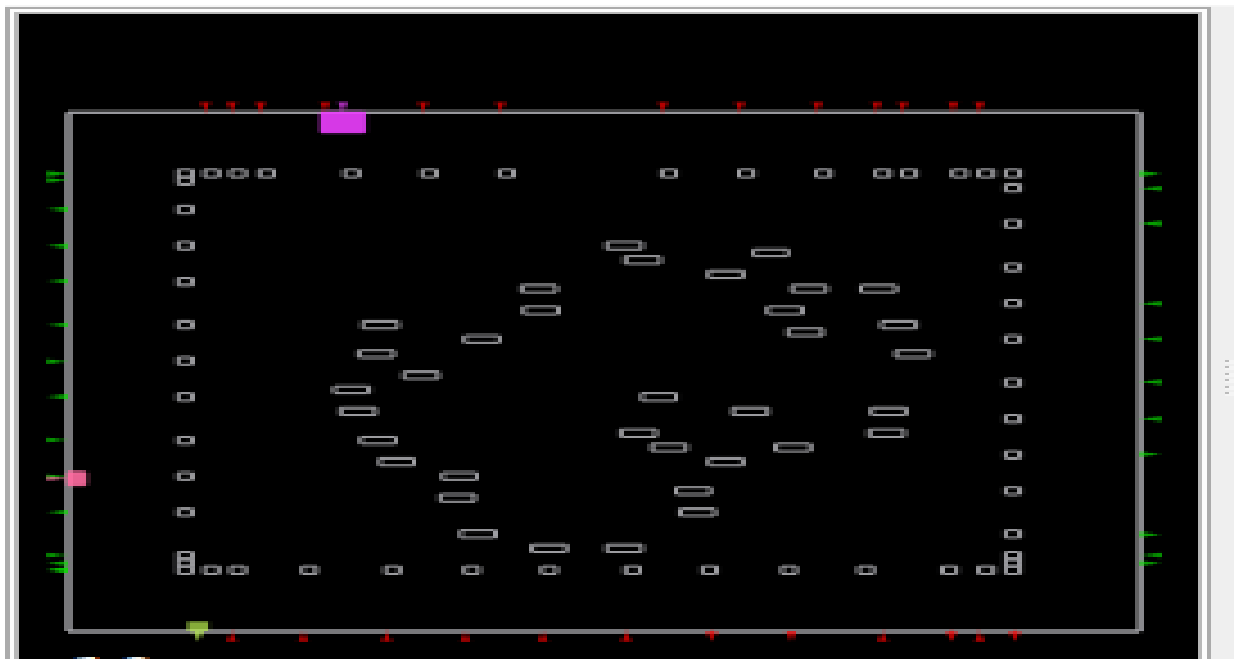
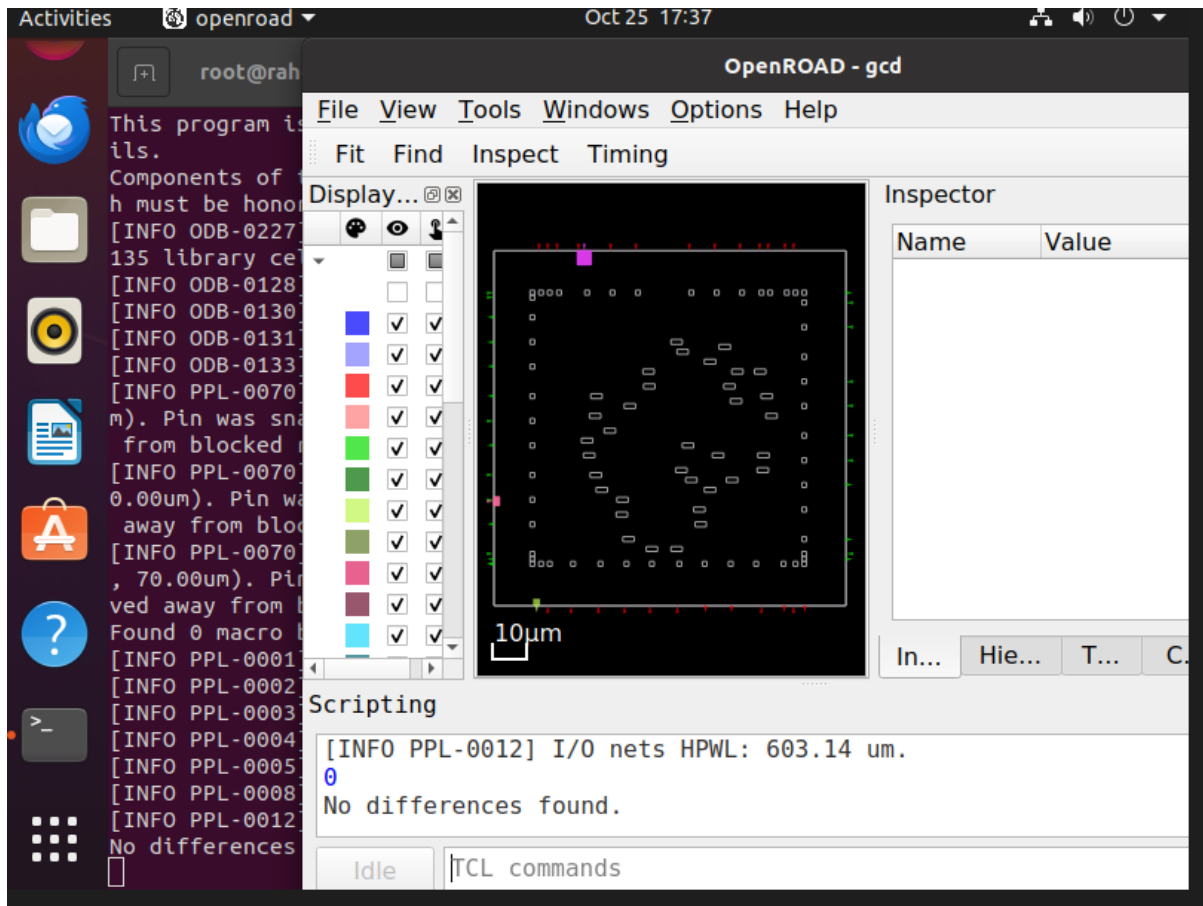
```

place_pin -pin_name clk -layer metal7 -location {40 30} -pin_size {1.6 2.5} -force_to_die_boundary
place_pin -pin_name resp_val -layer metal4 -location {12 50} -pin_size {2 2} -force_to_die_boundary
place_pin -pin_name req_msg[0] -layer metal10 -location {25 70} \
  -pin_size {4 4} -force_to_die_boundary
place_pins -hor_layers metal3 -ver_layers metal2 -corner_avoidance 0 -min_distance 0.12
set def_file [make_result_file place_pin4.def]

```

```
diff_file place_pin4.defok $def_file
```

```
source place_pin4.tcl
```



Macro or Standard Cell Placement

Macro Placement

In this section, you will explore various placement options for macros and standard cells and study the impact on area and timing.

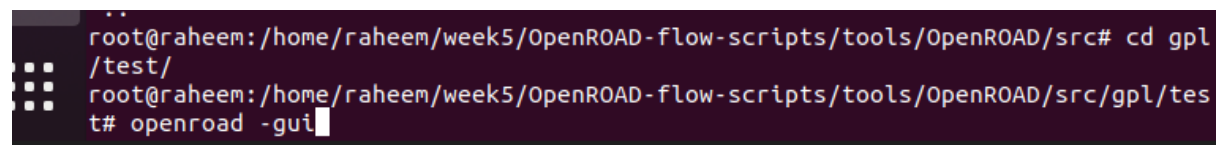
Refer to the following built-in example here to learn about macro placement.

```
source helpers.tcl
set test_name macro01
read_lef ./nangate45.lef
read_lef ./bp_be_top_macro.lef
read_def ./test_name.def
```

```
global_placement -density 0.7
set def_file [make_result_file $test_name.def]
write_def $def_file
diff_file $def_file $test_name.defok
source report_hpw.tcl
```

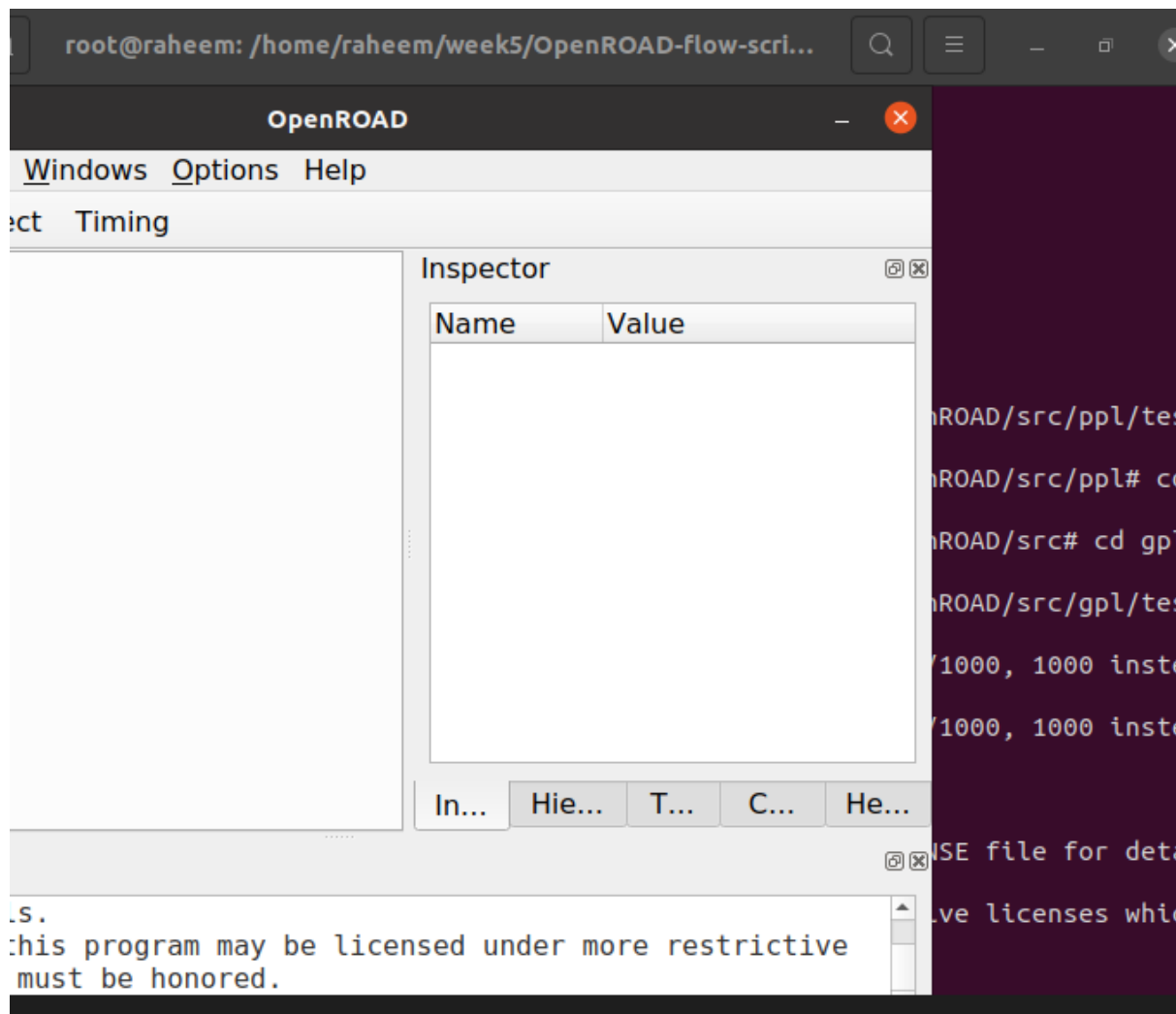
Placement density impacts how widely standard cells are placed in the core area. To view this in OpenROAD GUI run the following command(s) in the terminal in OpenROAD tool root directory:

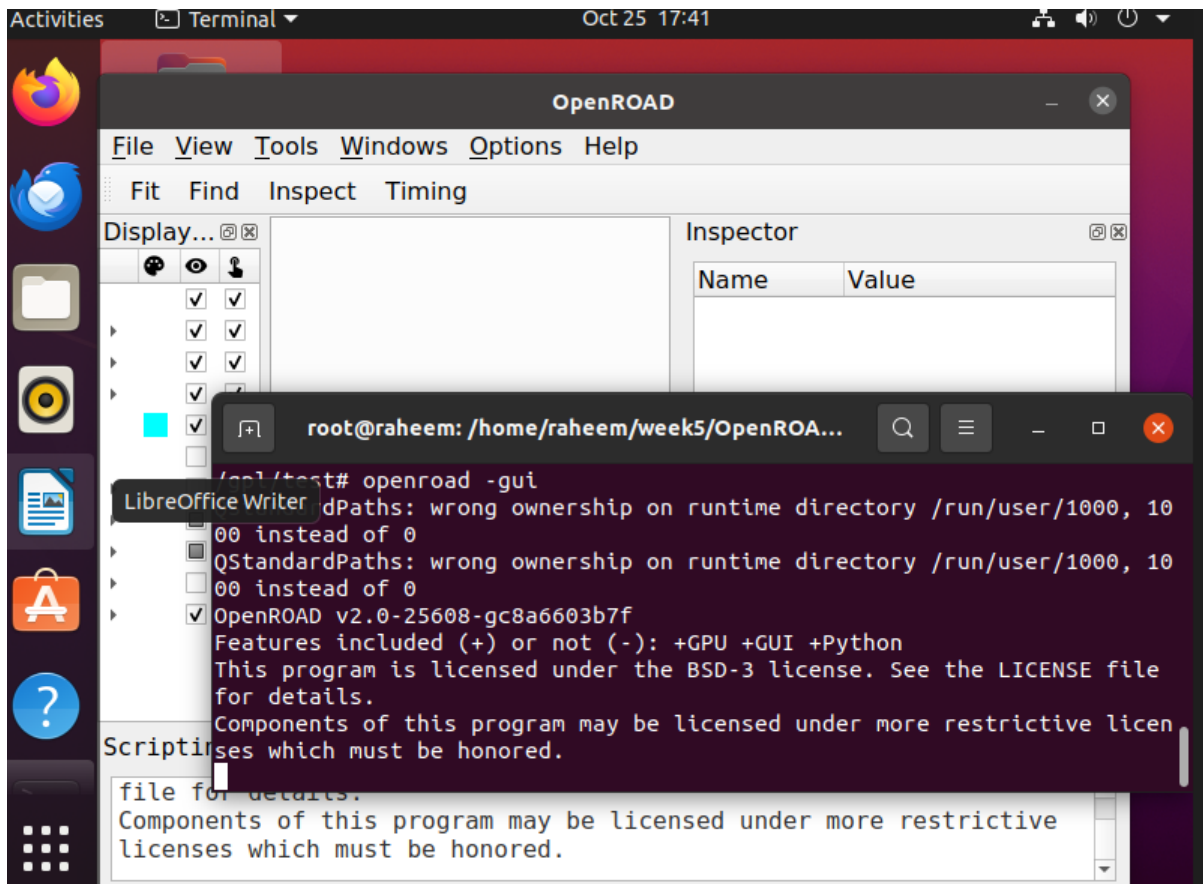
cd ../tools/OpenROAD/src/gpl/test/



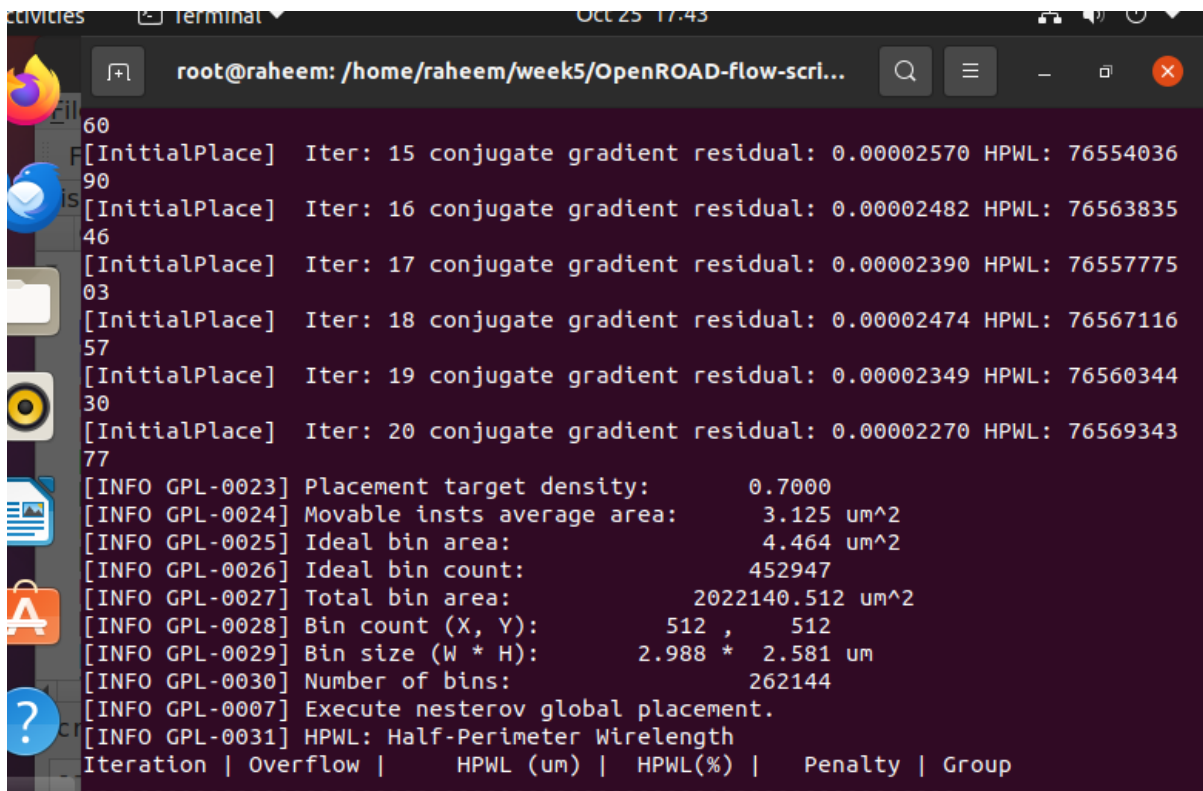
```
root@raheem:/home/raheem/week5/OpenROAD-flow-scripts/tools/OpenROAD/src# cd gpl
/test/
root@raheem:/home/raheem/week5/OpenROAD-flow-scripts/tools/OpenROAD/src/gpl/tes
t# openroad -gui
```

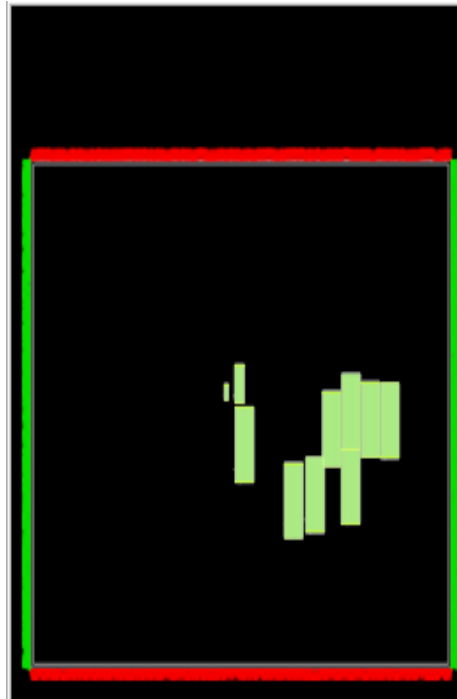
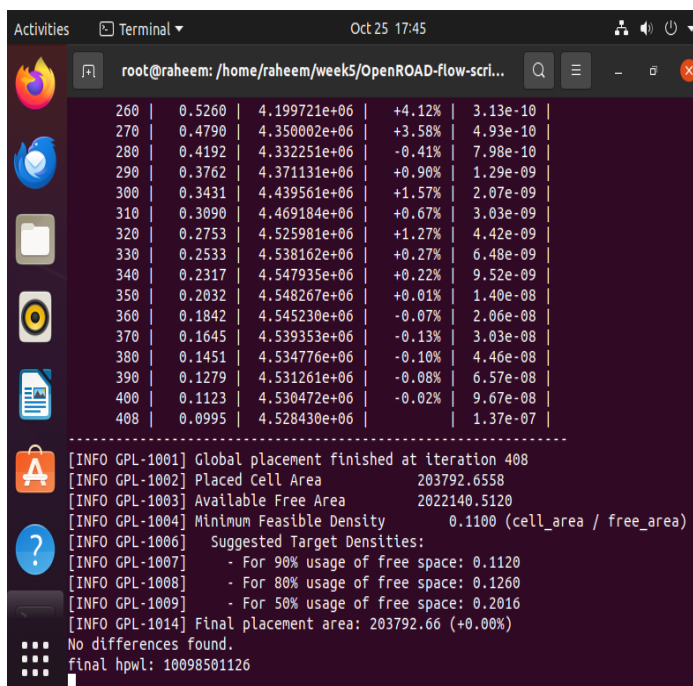
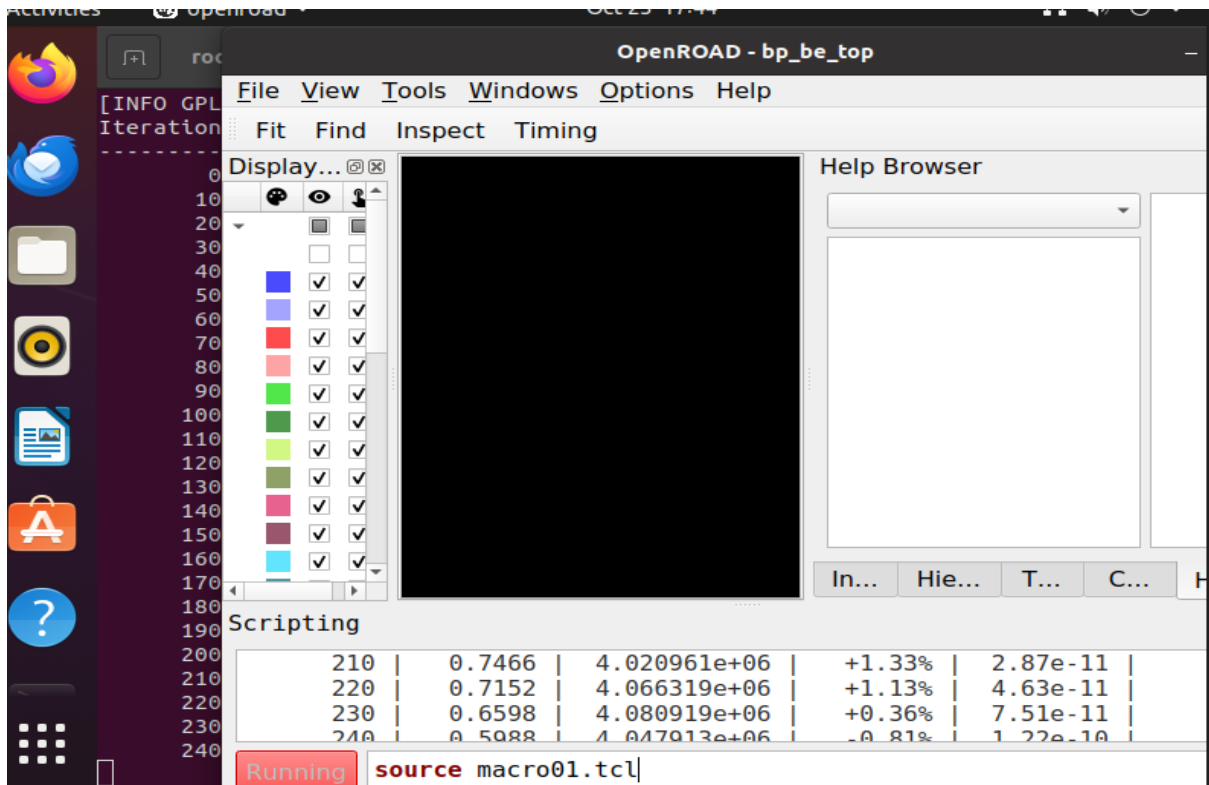
openroad -gui

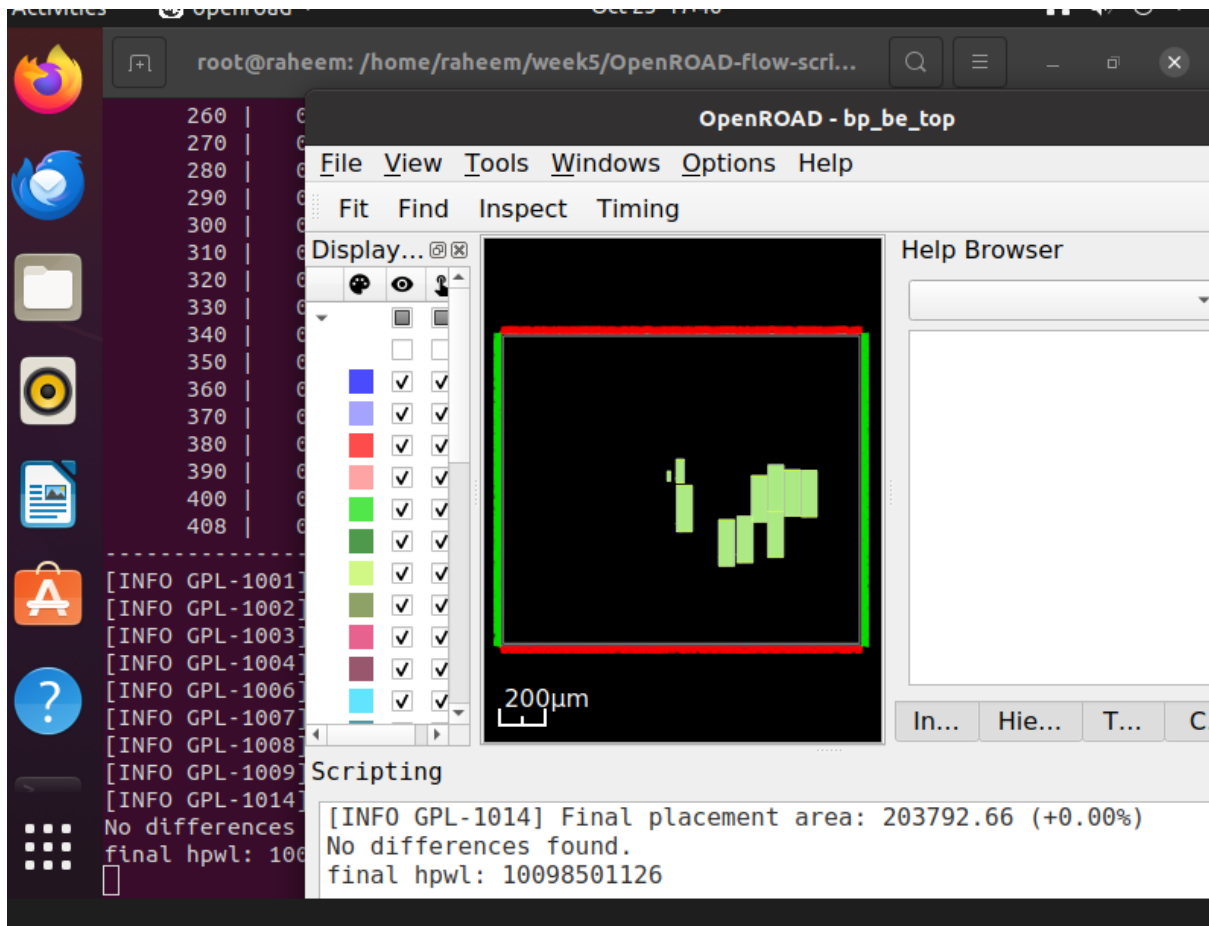




source macro01.tcl







So any issues!!! But Awesome in bugging last three days day and night 24 hours tried to sought every issue tried different machine ... every package made me to learn bison, or-tools, cudd, make, CMake, g++ compiler and finally nice stress... with debugging skills how to rm and build..

Journey of Yosys 0.9 to 0.58

After difficulties your efforts us relax!! Finally

Thank you for VSD Team for this RSIC V Tape-out Program Dr M A Raheem MJCET Hyderabad